

Final Report

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1 Introduction

Nuclear physics has been proposed as an important field of application for quantum computing, owing to the strong interactions that underlie nuclear dynamics. The complexity grows exponentially with the number of nucleons (protons or neutrons) present in the system, making classical calculations and simulations difficult. While classical ab initio calculations of nuclei have improved significantly over the last few decades [1] - even finding good results for nuclei with as many as 208 nucleons [2] - they are often limited to nuclear magic numbers [3] or calculations of properties, rather than dynamics. The exponential scaling of simulations outside these ideal cases motivates the use of quantum computing, which is better suited to the complex interactions between nucleons.

Dynamics within nuclei are determined by the underlying Quantum Chromodynamics (QCD) at the level of quarks and gluons, but these interactions are much too computationally intensive to include when considering large nuclei. At the low-energy scale of nuclear physics, Chiral Effective Field Theories (EFTs) have been developed that accurately capture the interactions between nucleons [4]. These theories include surprising terms like 3-body nucleon interactions, and long-range interactions are mediated by π mesons (pions). Computational methods for simulating field theories have advanced, including the important class of Lattice theories, which discretize space (and time) with a finite distance between points, greatly simplifying calculations. Quantum computing appears especially well-suited for simulating lattice theories due to the exponential scaling of system size with the number of qubits [5].

In this work, we consider Chiral EFTs on a lattice. Previous work by Watson et al. (2025) has investigated the cost of performing time evolution and Quantum Phase Estimation (QPE) for various system sizes using these theories [6]. They considered 3 EFTs, including Pionless EFT (where nucleons interact only through contact when occupying the same lattice site), One-Pion Exchange EFT (where long-range interactions are present, but pions are otherwise not included), and Dynamical Pion EFT (where pions are treated as a scalar field interacting with the nucleons). They performed resource estimation (including the number of logical qubits, T-gates, and two-qubit gates) for the tasks using Trotterization as a subroutine to implement the time-evolution operator. Motivated by algorithmic developments that have reduced costs for other applications ([7], [8]), we sought to reduce the logical resources required for simulating nuclear lattice EFTs. In our project, we implement Qubitization as the subroutine for the time evolution operator and compare the resource estimations for performing QPE for Dynamical Pion EFT.

2 Dynamical-Pion Effective Field Theory

Nuclear EFTs are the low-energy effective theories that describe the strong interactions between nucleons (protons and neutrons) bound inside atomic nuclei. Nucleons are spin- $\frac{1}{2}$ particles where the isospin I determines whether the nucleon is a proton ($I_z = \frac{1}{2}$) or a neutron ($I_z = -\frac{1}{2}$). Nucleon-nucleon interactions are mediated by pions with three flavors: π_+ , π_0 , and π_- . Dynamical-pion EFT explicitly includes pion degrees of freedom, allowing for pion propagation and nucleon-pion interactions beyond static one-pion exchange, making it a very powerful tool to study nuclear strong interactions.

A common approach to simulate nuclear EFT on a digital computer is to discretize space into evenly-spaced three-dimensional lattices [4]. The Pauli exclusion requires the nucleons in each site have to have different (S_z, I_z) number, limiting the maximum number of nucleons per site to four. From here, we will describe the spin-isospin as $\sigma = (S_z, I_z)$ where $S_z = \pm\frac{1}{2}$ and $I_z = \pm\frac{1}{2}$.

The $D\pi$ Hamiltonian in general can be described in Equation 1 [6]

$$H_{D\pi} = H_{\text{free}} + H_C + H_{C_{I^2}} + H_\pi + H_{N\pi} \quad (1)$$

The free term H_{free} describes the kinetic energy of the nucleons. On a lattice, this term corresponds to the hopping of nucleons between neighboring sites $\langle \mathbf{x}, \mathbf{y} \rangle$.

$$H_{\text{free}} = -h \sum_{\langle \mathbf{x}, \mathbf{y} \rangle} \sum_{\sigma} \left(a_{\sigma}^{\dagger}(\mathbf{x}) a_{\sigma}(\mathbf{y}) + a_{\sigma}^{\dagger}(\mathbf{y}) a_{\sigma}(\mathbf{x}) \right) + 6h \sum_{\mathbf{x}} \sum_{\sigma} N_{\sigma}(\mathbf{x}) \quad (2)$$

where $h = \frac{1}{2Ma_L^2}$ denotes the overall scaling of the kinetic energy, a_L is distance between lattice sites, $a^{\dagger}$ and a are the creation and annihilation operators, and $N = a^{\dagger}a$ is the number operator in the second quantized Hamiltonian. The contact terms H_C and $H_{C_{I^2}}$ describe short-range nucleon-nucleon interactions. On a lattice, it describes the interaction between the nucleons on the same lattice site $\mathbf{x}$

$$H_C = \frac{C}{2} \sum_{\mathbf{x}} : \left(\sum_{\alpha} \sum_{\beta} a_{\alpha\beta}^{\dagger}(\mathbf{x}) a_{\alpha\beta}(\mathbf{x}) \right)^2 : \quad (3)$$

$$H_{C_{I^2}} = \frac{C_{I^2}}{2} \sum_{\mathbf{x}} \sum_I : \left(\sum_{\alpha} \sum_{\beta, \delta} a_{\alpha\beta}^{\dagger}(\mathbf{x}) [\tau_I]_{\beta\delta} a_{\alpha\delta}(\mathbf{x}) \right)^2 : \quad (4)$$

where $\alpha\beta$ denotes spin of α and isospin of β , τ_I with $I \in \{1, 2, 3\}$ are the Pauli matrices acting on isospin space, and $:\hat{O}:$ specifies the normal ordered form of $\hat{O}$. The constants C and C_I are specified later in Table 1 and are chosen to match those used in Ref. [6].

The free pion dynamics are represented by H_π

$$H_\pi = \frac{a_L^3}{2} \sum_{\mathbf{x}} \sum_I \left[\Pi_I^2(\mathbf{x}) + (\nabla \pi_I(\mathbf{x}))^2 + m_\pi^2 \pi_I^2(\mathbf{x}) \right] \quad (5)$$

where π_I is the bosonic scalar pion field, Π_I is its conjugate momentum field for isospin indices $I \in \{1, 2, 3\}$, and ∇ is the finite difference derivative. The pion-nucleon interaction is described by $H_{N\pi}$, which is the sum of the axial-vector term, H_{AV} , and the Weinberg-Tomozawa term H_{WT}

$$H_{N\pi} = H_{AV} + H_{WT} \quad (6)$$

where

$$H_{AV} = \frac{g_A}{2f_\pi} \sum_{\mathbf{x}} \sum_{\alpha, \beta, \gamma, \delta} \sum_{I, S} a_{\alpha\beta}^\dagger(\mathbf{x}) [\tau_I]_{\beta\delta} [\sigma_S]_{\alpha\gamma} \partial_S \pi_I(\mathbf{x}) a_{\gamma\delta}(\mathbf{x}) \quad (7)$$

$$H_{WT} = \frac{1}{4f_\pi^2} \sum_{\mathbf{x}} \sum_{I_1, I_2, I_3} \sum_{\alpha, \beta, \delta} \epsilon_{I_1 I_2 I_3} \pi_{I_2}(\mathbf{x}) \Pi_{I_3}(\mathbf{x}) a_{\alpha\beta}^\dagger(\mathbf{x}) [\tau_{I_1}]_{\beta\delta} a_{\alpha\delta}(\mathbf{x}) \quad (8)$$

where σ_S with $S \in \{1, 2, 3\}$ are the Pauli matrices acting on spin space.

3 Methods

3.1 Hamiltonian Encodings and parameters

We chose a three-dimensional cubic lattice with L sites per side to represent the Dynamical Pion EFT Hamiltonian $H_{D\pi}$. All the terms are constructed with `OpenFermion`[9]. Each fermionic term is also transformed into a Pauli string using the Jordan-Wigner transformation. All the physical parameters used in our simulations matched those used in [6]. The descriptions and values of those parameters are described in Table 1. Those Hamiltonian encodings are inputs to our qubitization workflow described in subsection 3.2.

Symbol	Description	Value
m_N	Nucleon mass	938 MeV
m_π	Pion mass	135 MeV
g_A	Axial coupling constant	1.26
f_π	Pion decay constant	93 MeV
C	Contact interaction coefficient	-51.94 MeV
C_{I^2}	Isospin-dependent contact coefficient	1.73 MeV
a_L	Lattice spacing	2.2 fm
h	Hopping coefficient	(derived from m_N, a_L)

Table 1: Parameters used in the dynamical-pion EFT Hamiltonian. Parameters are chosen to match those in [6].

We note that each site can hold at most four nucleons, so we need four qubits per site to encode the nucleon degree of freedom. Pions, however, are bosons, and the bosonic encoding requires a far larger number of qubits. We make use of the results in Lemma 5 of Watson et al. [6], which gives an estimate on the number of qubits needed to encode the pion degree of freedom to achieve a wavefunction overlap within a level of accuracy. This number, denoted as n_b , depends on the cut-off precision ϵ_{cut} , lattice size L , the number of nucleons A , and other physical parameters. Moreover, there are three flavors of pions, making the number of qubits needed to encode the pion degree of freedom is $3n_b$. On our 3D lattice, the total of qubits needed to encode the whole Hamiltonian is $L^3(4 + 3n_b)$.

The Hamiltonian includes the free-pion term, which accounts for the energy of the scalar pion field. This term has both position and momentum representations of the field (π and Π), which are Fourier Transform pairs of one another. Rather than represent the entire Hamiltonian in one space, each term is diagonalized in its own subspace, and the Quantum Fourier Transform (QFT) is applied when switching between the spaces. The pion operator terms are not fermionic, so instead of using the Jordan-Wigner Transformation, their encoding comes directly from the work of Ref. [6].

3.2 Qubitization Workflow

Qubitization is the process of constructing the Szegedy quantum walk operator (henceforth called the “walk operator” or “Qubitized walk operator”) W_H associated with the Hamiltonian H . The walk operator is constructed by first writing the Hamiltonian as a Linear Combination of Unitaries (LCU), which usually means a linear combination of Pauli strings. W_H is made by combining a so-called “prepare oracle” and “select oracle,” along with the Qubitization protocol. The walk operator is a Qubitized block encoding of the Hamiltonian, which, upon repeated application, implements polynomial powers of H . This allows for the construction of an approximation to the time evolution operator e^{-iHt} using an algorithm called Quantum Signal Processing (QSP) [10]. For a complete pedagogical discussion of this process, see Section II of Ref. [5].

In our project, we construct the Hamiltonians with `OpenFermion` and the encodings for the pion terms as given in Ref. [6] according to input parameters, including the lattice size L and number of nucleons A . We used a lattice spacing $a_L \approx 2.2$ fm, which results in parameters $h = 4.29$ MeV, $C = -51.94$ MeV, and $C_\pi = 1.73$ MeV. These Hamiltonians are fed into `pyLIQTR` as inputs to generate the Qubitized walk operator. We then run `estimate_resource` to obtain the number of logical T -gates needed to implement each walk operator. See the subsection 3.4 for more information. We chose lattice sizes of $L = 2, 3, 4$ due to limitations of computational resources. For each L , we vary the number of nucleons A from 1 to 100. In terms of constructing the Hamiltonian, the number of nucleons only affects the pion field cutoff, and thus the number of qubits n_b . Note that although we can only physically hold at most 32 nucleons in the $2 \times 2 \times 2$ lattice, we can still construct the Hamiltonian with the necessary n_b according to Lemma 5 of Ref. [6] and perform resource estimation. Although we could not prepare a state with this many nucleons present in the system (full occupation is at most $4 * L^3$ nucleons), we still keep those data points in our results for reference to better understand resource scaling. True applications of Dynamical Pion EFT are in the limit of large lattice sizes (which allow for better resolution and simulation accuracy), so the limit on the number of nucleons is not a problem with the model.

In the Qubitization workflow, the pion field momentum operator Π is written as a diagonal matrix, but it cannot be diagonalized simultaneously with π . Thus, we break the Hamiltonian into two terms: one in position space (π) and one in momentum space (Π), and we generate two separate walk operators for the two spaces. We estimate the resources for the QFT between the spaces and include them in the results, despite their small contribution to the overall cost. This is the same process as researchers would use in a full QPE simulation, where applying the QFT between operators in different spaces helps reduce resources.

For a Hamiltonian with spectral norm λ (the sum of absolute values of coefficients in an LCU expansion), performing QPE to an energy accuracy ΔE requires at most

$$N_{\text{walk}} = \frac{\sqrt{2}\pi\lambda}{\Delta E} \quad (9)$$

calls to the walk operator [11]. We used this result to estimate the resources needed to perform Qubitization-based QPE at accuracy $\Delta E = 1$ MeV. The results are shown and discussed in section 4.

3.3 Trotterization Bounds

For comparison, we estimate the resources needed to implement the time evolution of the dynamical-pion EFT hamiltonian with Trotterization. We use the characteristic time T_{cross} de-

fixed as the time for a nucleon to cross the lattice [6]

$$T_{\text{cross}} = a_L L \sqrt{\frac{M_N}{2E_{\text{kin}}}} \quad (10)$$

In our estimation, $a_L = 2.2 \text{ fm}$, $E_{\text{kin}} = 10 \text{ MeV}$, $M_N = 938 \text{ MeV}$, matching the values used by Watson et al. [6].

We make use of the result of Lemma 23 from Watson et al. [6], which stated that the expected T -gates count required to implement a first-order Trotter step of the time evolution of $H_{D\pi}$ within precision δ is

$$g(n_b, L)(1.15 \log(2g(L, n_b)/\delta) + 9.2) \quad (11)$$

where $g(n_b, L) = (45n_b^2 + 114n_b + 76)L^3$. To estimate δ needed for a total precision ϵ_{tot} , we make use of the fault-tolerance error budget stated in Table VII of Watson et al. [6], which implies that $\delta = \epsilon_{\text{tot}}/3r$ where r is the number of Trotter steps. The value of r is estimated from first-order Trotterization coarse bound in the same paper

$$\|U_{\text{Trot.}} - (e^{-iH_{D\pi}T_{\text{cross}}/r})^r\| \leq C_1 \Pi_{\text{max}}^2 \pi_{\text{max}}^2 L^3 \frac{T_{\text{cross}}^2}{r} \quad (12)$$

where $\Pi_{\text{max}}, \pi_{\text{max}}$ are the pion cutoffs that is a part of Lemma 5 in Watson et al. [6] and both of them are of $\mathcal{O}(\sqrt{AL^3E/\epsilon_{\text{cut}}})$. Setting the left-hand side to be the total error ϵ_{tot} , we get

$$r \leq r_{\text{max}} = C_1 \Pi_{\text{max}}^2 \pi_{\text{max}}^2 L^3 \frac{T_{\text{cross}}^2}{\epsilon_{\text{tot}}} \quad (13)$$

The number of T -gates for a time evolution of the dynamical-pion EFT hamiltonian estimated from our calculation is then

$$N_T = r_{\text{max}} [g(n_b, L)(1.15 \log(2g(L, n_b)/(\epsilon_{\text{tot}}/3r_{\text{max}})) + 9.2)] \quad (14)$$

In our calculation, we set $\epsilon_{\text{tot}} = 1$. The constant C_1 , which is independent of L , is heuristically determined by fitting our calculations to the data presented in Watson et al. [6], where they explicitly calculated the Trotter errors for $L = 10$.

3.4 Data and Software

Our code lives in the above github repository, NuQu (see [12]). Core features include Hamiltonian construction, resource estimation through pyLIQTR, and data collection and analysis. Our Hamiltonian construction works for 1-, 2-, and 3-dimensional lattices.

When sweeping over a range of A from 1 to 100, we generated the Hamiltonian according to the lattice size L and n_b calculated to be sufficient for the given A . After estimating resources using the estimation tools, the data is saved into a data folder in the repository, and can be plotted in various ways. The most interesting resource for this project is the number of T -gates required to implement the full QPE using the walk operator. A few plots are shown in section 4, including T -gate cost vs A for various values of L , and T -gate cost vs L for various values of A . These plots show the comparison between resource estimations for Qubitization (computed with pyLIQTR in this work) and Trotterization (using equation 14 from Ref. [6]). The results can be reproduced by rerunning the nucleon sweep with parameters matching ours. Instructions for reproduction are in the README in the repository.

While performing resource estimation for the Qubitized workflow, we also save other results, including the number of logical qubits required to implement the walk operator, the

runtime for Hamiltonian construction and resource estimation, the spectral norm λ , Clifford operations, and T-gate cost due to QFT applications. This data can be used both for further analysis in the project and also for improving the code and model. Specifically, the wall clock runtime of the code is important for understanding computational limitations and finding where optimization could help us find better results by estimating resources for larger lattice sizes.

4 Results

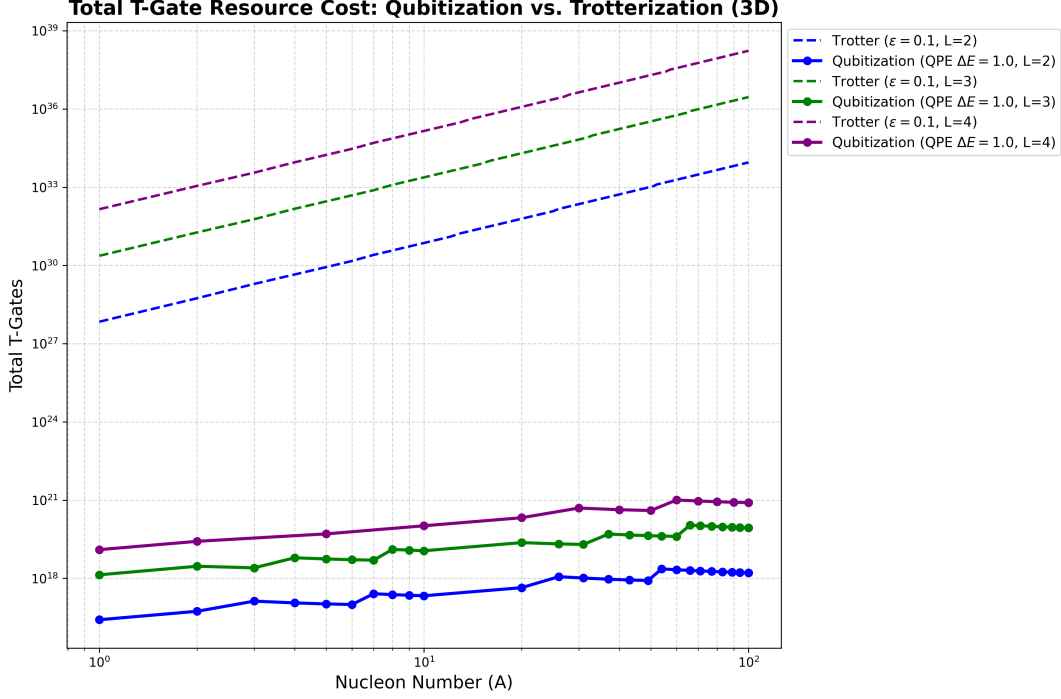


Figure 1: T-gate cost for full QPE vs Nucleon Number for Qubitized QPE (this work) and Trotterized Time evolution ([6]). $L = 2, 3$, and 4 lattice sites per side of a cubic lattice.

The results for 2, 3, and 4 lattice sites per side ($L = 2$, $L = 3$, and $L = 4$) are shown in Figure 1. The runtime for our code grew quickly with L , so we do not have results for larger lattice sizes due to the computational cost of constructing the Hamiltonian and estimating resources. We find that Qubitized QPE requires around 10 orders of magnitude less than Trotterized QPE at the cheapest, and it scales much better with A .

It is also of interest to plot the number of required T-gates against the lattice size L as shown in Figure 2. On the log-log plots, the Qubitized resource estimates have a smaller slope than the Trotterized T-gate cost, which indicates that it scales better with L . This is an important result, as simulations in lattice theories are usually checked by taking the lattice spacing a_L to be small, and L large, so that the lattice is a closer approximation to a continuum of real space. Future simulations on quantum computers using these Chiral EFTs on a lattice will require large lattices to confirm the accuracy of their results.

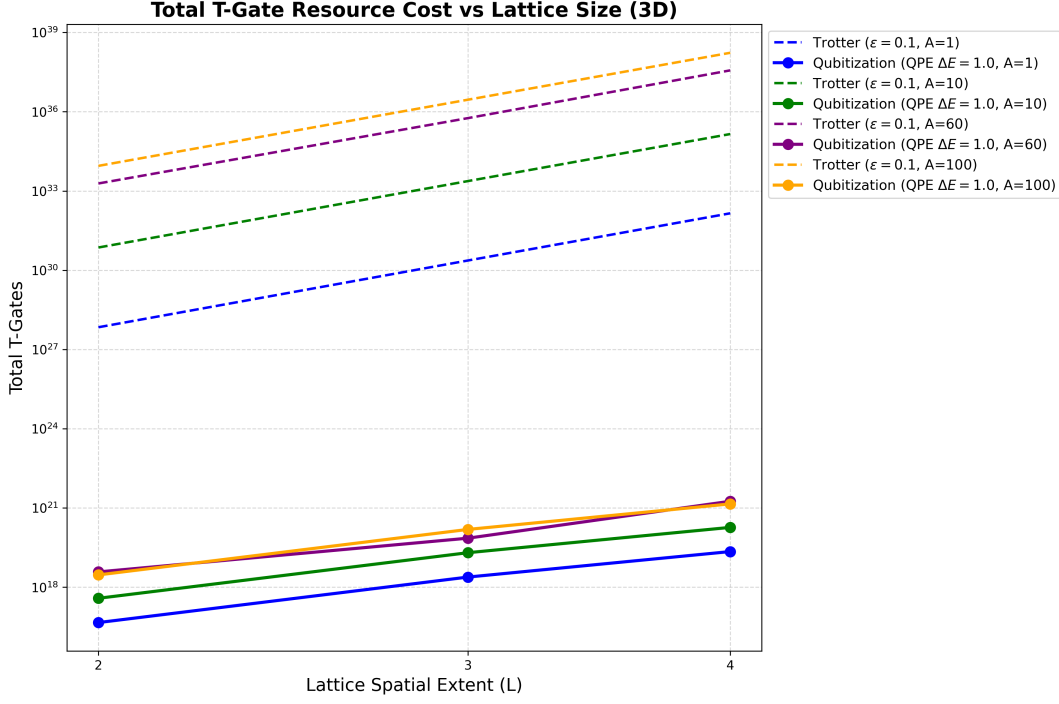


Figure 2: T-gate cost for full QPE vs Lattice size L for Qubitized QPE (this work) and Trotterized QPE ([6]). Results for $A = 1, 10, 60$ and 100 nucleons present in the model are shown.

5 Resource Implications

As classical computing resources and methods continue to scale and improve, the space where quantum computing will have an advantage shifts towards more complex systems. Nuclear physics appears to be a promising application of quantum computing due to the strong interactions and correlations between nucleons. Previous works investigating applications, including stopping power calculations for inertial confinement fusion [13], Lattice Quantum Chromodynamics [14], and high-energy physics [15], all report high resource costs infeasible in the near future. Driving down the resource estimates will help us find useful applications for early fault-tolerant quantum computing, which requires algorithmic development and implementation into resource estimates.

In this work, we estimated the number of T -gates needed to perform qubitization-based QPE on the Dynamical Pion Hamiltonian. Comparing to the first estimates from [6] which estimated the resource costs for time evolution and QPE using Trotterization, we find that Qubitization appears to have much lower costs. This result is important for comparing the necessary resources between the two algorithms for nuclear lattice EFTs. Different physical systems require different approaches to solve, so displaying the decrease in costs using Qubitization instead of Trotterization is an important step in finding useful applications for early fault-tolerant quantum computing within nuclear physics.

6 Limitations and future work

The biggest limitation of this project so far has been the lattice size L . Watson et al. [6] presented results for $L = 10$, so our comparison at small lattice sizes is not as strong for arguing reduced costs. We plan to optimize the code to construct the Hamiltonians and estimate resources more efficiently, and move to a high-performance computing cluster to run for much

larger lattices. We hope to find that the resource scaling with L continues to be favorable for Qubitization, which would indicate that the nuclear lattice EFTs could be a useful application of quantum computing in the future.

Additionally, although we used the scaling formulas for the Trotter cost from Ref. [6], the error bounds were left in big-O notation. As we mentioned in section 3.3, we fit the constant C_1 heuristically to match their full T-gate cost estimation for $L = 10$, but this needs a further validation check. We want to further understand the error formulas and reproduce the resource estimates from the other paper to verify our comparison and the huge improvement gained by Qubitization.

We expect that the reduced costs for using Qubitization will come at the cost of requiring more ancilla qubits to implement the time evolution operator. Unfortunately, while wrapping up the project, we discovered that our code did not save the logical qubit counts required for implementing the walk operator. We plan to fix the code and quantify in future work how many logical ancillae are required to implement Qubitization. In the era of early fault-tolerant quantum computing, there will be a tradeoff between the number of qubits and the runtime, so quantifying the qubit cost for implementing EFTs is an important part of demonstrating their applicability.

One important note is that Ref. [6] utilizes the Verstraete-Cirac (VC) encoding [16] as opposed to the Jordan-Wigner transformation for converting fermionic operators into Pauli strings. They claim that this reduces the Pauli weight of the terms in the Hamiltonian, so another step in the project is to implement the VC encoding in our code and compare the results. Within the nuclear physics community, the effect of these encodings has not been studied very much, but the local nature of the VC encoding seems likely to reduce resources. At first glance, it might also take advantage of the properties of the Hamiltonian, so this investigation is worth pursuing.

Taking inspiration from other application fields such as quantum chemistry, we also hope to implement recent algorithmic developments, including double factorization, tensor hypercontraction [7] and its derivatives [8] to further reduce the logical resources required for nuclear lattice EFTs. Much progress has been made in these other fields, and one of our goals is to show that nuclear physics is an equally important and viable application for early fault-tolerant quantum computing. We recognize that there are both similarities and differences between electronic and nuclear systems, which likely require different approaches to algorithmic development. We hope to demonstrate methods that generalize beyond quantum chemistry and to discover new algorithms specialized for nuclei.

Finally, we want to make our resource estimates more physical by calculating a physical runtime for the number of T-gates we computed in the project. This will require making architectural decisions such as choosing the qubit modality, error correcting code, and the number of T-state factories in the quantum computer. This encourages research and collaboration across more parts of the full stack of quantum computing, and will help ground the result in a meaningful runtime for a useful application.

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